

Probability
Chapter 2 - Module 1

Real Valued Random Variables

Richard Redner

June 21, 2026

Chapter 1 was focused on the sample space, outcomes and events. Most of the examples were finite sample spaces and the probability arguments were simple counting problems.

We also introduced the probability axioms and derived a few simple theorems.

We now move to, for what to me is more exciting territory, which is the study of **Random Variables**. This will remain the focus of the course from here on out, although we will still need sample spaces, events, etc. and all of the material from Chapter 1.

Example 1. Suppose that we once again roll two fair dice and we measure the sum of the two dice.

So once again the sample space is

$$S = \{(i, j) | 1 \leq i, j \leq 6\}$$

and we define a function

$$X : S \rightarrow \{1, 2, 3, \dots, 11, 12\}$$

by $X(i, j) = i + j$ so that the random variable is the sum of the two dice.

Then by simple counting

$$P(X = 1) = 0$$

$$P(X = 2) = P(X = 12) = 1/36$$

$$P(X = 3) = P(X = 11) = 2/36$$

$$P(X = 4) = P(X = 10) = 3/36$$

$$P(X = 5) = P(X = 9) = 4/36$$

$$P(X = 6) = P(X = 8) = 5/36$$

$$P(X = 7) = 6/36$$

Example 2. Let p be the probability that a coin comes up H . Let N be the number of flips of the coin until the coin comes up H .

Question. For $k = 1, 2, \dots$, what is $P(N = k)$.

By the definition of p , $P(N = 1) = p$.

To continue, define C_i as the outcome of the i^{th} flip of the coin so that $P(C_i = H) = p$ and $P(C_i = T) = 1 - p$ for each integer i .

So $N = 2$ only when $C_1 = T$ and $C_2 = H$. Since these are independent events,

$$\begin{aligned} P(N = 2) &= P(C_1 = T \wedge C_2 = H) \\ &= P(C_1 = T) P(C_2 = H) \\ &= P(N \neq 1) P(C_2 = H) \\ &= (1 - p)p. \end{aligned}$$

We note that $P(N > 1) = 1 - p$. In general one can show using induction that

$$P(N > k) = (1 - p)^k$$

and use this to verify that

$$\begin{aligned} P(N = k + 1) &= P(N > k \wedge C_{k+1} = H) \\ &= P(N > k) P(C_{k+1} = H) \\ &= (1 - p)^k p \end{aligned}$$

for $k = 1, 2, \dots$

So for each $k = 1, 2, \dots$,

$$P(N = k) = p(1 - p)^{k-1}.$$

We should check that the sum of these events is one. Assuming that $p \neq 0$ we can see that

$$\begin{aligned}\sum_{k=1}^{\infty} P(N = k) &= \sum_{k=1}^{\infty} p(1-p)^{k-1} \\ &= p \sum_{j=0}^{\infty} (1-p)^j \\ &= p \frac{1}{1-(1-p)} = 1,\end{aligned}$$

where we have let $j = k - 1$ we have used the fact that for $|a| < 1$ that

$$\sum_{j=0}^{\infty} a^j = \frac{1}{1-a}.$$

Note that since $P(N = k) = p(1-p)^{k-1}$ that by definition (section 2.23, page 28), N in Example 2. is a **geometric** random variable.

Def. (Real Random Variable) Given a sample space S , a **Real Valued Random Variable** is a (measurable) function

$$X : S \rightarrow \mathcal{R}.$$

Def. (Discrete Random Variable) Let X be a real valued random variable that takes on either a finite number of values, say $x_1, x_2, \dots, x_M$ or a countably infinite number of values $x_1, x_2, \dots$ for which $P(X = x_i) > 0$ for each i .

Such a random variable is called a **Discrete** random variable and if $P(X = x_i) > 0$, then x_i is called a mass point.

If we then define $p(x_i) = P(X = x_i)$, then $\sum_{p(x_i) > 0} p(x_i) = 1$.

And let's note that in all cases if $x_i \neq x_j$ that

$$P(X = x_i \vee X = x_j) = P(X = x_i) + P(X = x_j).$$

Why? - It is important that you justify this.

Def. (CDF) For a real random variable X , for any real number a the cumulative distribution function (CDF) is

$$F(a) = P(X \leq a)$$

Example 1. (revisited)

$$\begin{aligned} F(4) &= P(X \leq 4) \\ &= P(X = 1, 2, 3 \text{ or } 4) \\ &= P(X = 1) + P(X = 2) + P(X = 3) + P(X = 4) \\ &= \frac{0 + 1 + 2 + 3}{36} = \frac{6}{36} \end{aligned}$$

and similarly $F(3.99) = P(X \leq 3.99) = P(X \leq 3) = \frac{3}{36}$.

Theorem. Let F be a cumulative distribution function for a real random variable. Then

1. F is non-decreasing.
2. $\lim_{x \rightarrow \infty} F(x) = 1$
3. $\lim_{x \rightarrow -\infty} F(x) = 0$
4. For a discrete random variable, F has a jump discontinuity at each mass point x_i .
5. F is right continuous, which is to say that $\lim_{x \rightarrow a^+} F(x) = F(a)$.

Def. (nondecreasing) A function $g(x)$ is said to be nondecreasing if for each x_1 and x_2 ,

$$x_1 \leq x_2 \text{ implies that } g(x_1) \leq g(x_2).$$

Example 2. (revisited) Suppose the we flip a coin 10 times (not necessarily a fair coin) with probability of success p (we get a H). What is the probability that we get an H within the first ten tries. The answer is $F(10)$, where

$$\begin{aligned}
 F(10) &= P(N \leq 10) \\
 &= \sum_{k=1}^{10} P(N = k) \\
 &= \sum_{k=1}^{10} p(1-p)^{k-1} \\
 &= p \sum_{j=0}^9 (1-p)^j \\
 &= p \frac{1 - (1-p)^{10}}{1 - (1-p)} \\
 &= 1 - (1-p)^{10},
 \end{aligned}$$

since $\sum_{j=0}^n a^j = \frac{1 - a^{n+1}}{1 - a}$ as long as $a \neq 1$.

Def. (Binomial Random Variables) Given n independent trials with probability of “success” p and “failure” $(1 - p)$, let X denote the number of successes. Then X is a **binomial** random variable and

$$P(X = i) = \binom{n}{i} p^i (1 - p)^{n-i}$$

for $i = 0, 1, \dots, n$ and where

$$\binom{n}{i} = \frac{n!}{i! (n - i)!}.$$

Note that

$$(a + b)^n = \sum_{i=0}^n \binom{n}{i} a^i b^{n-i}$$

and so

$$\sum_{i=0}^n P(X = i) = \sum_{i=0}^n \binom{n}{i} p^i (1 - p)^{n-i} = (p + (1 - p))^n = 1^n = 1.$$

Question: If we flip a fair coin 20 times what is the probability that we get 10 H and 10 T?

Answer:

Notationally the solution is $p_n(10)$ for the binomial random variable with $n = 20$ and $p = 1/2$ and its value is

$$p_n(10) = \binom{20}{10} \left(\frac{1}{2}\right)^{10} \left(\frac{1}{2}\right)^{20-10} = \binom{20}{10} \frac{1}{2^{20}}.$$

Question. Is $p_n(10) = 1/2$, $< 1/2$, $> 1/2$, $\approx 1/2$.

Def. (Poisson Random Variable) A random variable X that takes on the values $0, 1, \dots$ is said to be a **Poisson** random variable with parameter $\lambda > 0$ if

$$p(i) = P(X = i) = e^{-\lambda} \frac{\lambda^i}{i!}.$$

Note that

$$\sum_{i=0}^{\infty} p(i) = \sum_{i=0}^{\infty} e^{-\lambda} \frac{\lambda^i}{i!} = e^{-\lambda} \sum_{i=0}^{\infty} \frac{\lambda^i}{i!} = e^{-\lambda} e^{\lambda} = 1.$$

We will look at this example more closely later.

Def. (Discrete Expectations) Let X be a discrete random variable with mass function $p(x)$. Then

$$E[X] = \sum_{x:p(x)>0} x p(x).$$

Example 3. Let $X \sim b(n, p)$ a binomial random variable with parameters n and with probability of success p . Then

$$\begin{aligned} E[X] &= \sum_{i=0}^n i \binom{n}{i} p^i (1-p)^{n-i} \\ &= \sum_{i=1}^n i \frac{n!}{i!(n-i)!} p^i (1-p)^{n-i} \\ &= np \sum_{i=1}^n \frac{(n-1)!}{(i-1)!(n-i)!} p^{i-1} (1-p)^{n-i} \\ &= np \sum_{j=0}^{n-1} \frac{(n-1)!}{(j)!(n-1-j)!} p^j (1-p)^{n-1-j} \\ &= np. \end{aligned}$$

Def. (Continuous Random Variable) Let X be a real valued random variable. We say that X is a **continuous** random variable if there is a non-negative function f so that for $B \subset \mathcal{R}$,

$$P(X \in B) = \int_B f(x) dx$$

in which case we call f the probability density function for the continuous random variable X .

Truth in Advertising. Except in trivial cases ($f \equiv 0$ for example) it is actually not possible to define $\int_B f(x) dx$ for all B . This is true even if one considers Lebesgue integrals. Instead the integral is usually defined only on “measurable” subsets of $\mathcal{R}$.

We will not define “measurable” in this course, but it is likely that every subset of $\mathcal{R}$ that you might think of is measurable. So this is more of a theoretical than practical limitation and in the vast majority of cases that we will be interested in, B will be an interval.

Def. (Uniform Distribution) For any real numbers $a < b$, the uniform density function is

$$f(x) = \begin{cases} \frac{1}{b-a} & a \leq x \leq b \\ 0 & \text{otherwise.} \end{cases}$$

Def. (Normal Distribution) The normal density function is

$$f(x, \mu, \sigma) = \frac{1}{\sqrt{2\pi}} \frac{1}{\sigma} e^{-(x-\mu)^2/2\sigma^2}.$$

Def. (Expectation for a Continuous Random Variable) Let X be a continuous random variable with density function f . Then

$$E[X] = \int_{-\infty}^{\infty} x f(x) dx.$$

Example 4. Let X be normally distributed with mean μ and variance σ^2 ; $X \sim N(x, \mu, \sigma)$. Then for $y = \frac{x-\mu}{\sigma}$ we have that

$$\begin{aligned} E[X] &= \int_{-\infty}^{\infty} x \frac{1}{\sqrt{2\pi}} \frac{1}{\sigma} e^{-(x-\mu)^2/2\sigma^2} dx \\ &= \int_{-\infty}^{\infty} (\sigma y + \mu) \frac{1}{\sqrt{2\pi}} e^{-y^2/2} dy \\ &= \sigma \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} y e^{-y^2/2} dy + \mu \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-y^2/2} dy \\ &= \sigma \cdot 0 + \mu \cdot 1 = \mu. \end{aligned}$$

Def. Let X be a discrete random variable with probability mass function $p(x)$. And let $g(x) : \mathcal{R} \rightarrow \mathcal{R}$. Then

$$E[g(X)] = \sum_{x|p(x)>0} g(x)p(x).$$

Def. Let X be a continuous random variable with pdf $f(x)$. And let $g(x) : \mathcal{R} \rightarrow \mathcal{R}$. Then

$$E[g(X)] = \int_{-\infty}^{\infty} g(x)f(x) dx.$$

The most important example is probably $g(x) = (x - \mu)^2$, where $\mu = E[X]$.

Def. (Variance) The variance of a random variable is

$$\text{Var}(X) = E[(X - E[X])^2].$$

Example 4b. Let X be a normally distributed random variable $X \sim N(x, \mu, \sigma)$. Then

$$\text{Var}(X) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} (x - \mu)^2 \frac{1}{\sigma} e^{-(x-\mu)^2/2\sigma^2} dx = \sigma^2.$$

A Useful Fact: Let a and b be real numbers and X a random variable. Then

$$E[aX + b] = aE[X] + b.$$

We prove this for the case that X is a real random variable with pdf $f(x)$, in which case

$$\begin{aligned} E[aX + b] &= \int_{-\infty}^{\infty} (ax + b) f(x) dx \\ &= a \int_{-\infty}^{\infty} x f(x) dx + b \int_{-\infty}^{\infty} f(x) dx \\ &= a E[X] + b. \end{aligned}$$

Another Useful Fact:

For any real random variable X , let $\mu = E[X]$. Then using linearity

$$\begin{aligned}\text{Var}(x) &= E[(X - E[X])^2] \\&= E[X^2 - 2XE[X] + E[X]^2] \\&= E(X^2 - 2X\mu + \mu^2) \\&= E[X^2] - E[2\mu X] + E[\mu^2] \\&= E[X^2] - 2\mu E[X] + E[\mu^2] \\&= E[X^2] - 2\mu\mu + \mu^2 \\&= E[X^2] - \mu^2 \\&= E[X^2] - E[X]^2.\end{aligned}$$

Example 2. (revisited) Suppose that $P(N = k) = p(1 - p)^{k-1}$.

To compute $E[N]$ we first note that since

$$g(x) = \frac{1}{1-x} = \sum_{k=0}^{\infty} x^k \quad \text{for } |x| < 1$$

then

$$g'(x) = \frac{1}{(1-x)^2} = \sum_{k=1}^{\infty} kx^{k-1} \quad \text{for } |x| < 1.$$

So

$$\begin{aligned} E[N] &= \sum_{k=1}^{\infty} k P(N = k) \\ &= \sum_{k=1}^{\infty} k p(1-p)^{k-1} \\ &= p \sum_{k=1}^{\infty} k(1-p)^{k-1} \\ &= p \frac{1}{(1-(1-p))^2} = \frac{p}{p^2} = \frac{1}{p}. \end{aligned}$$

We also note that

$$g''(x) = \frac{2}{(1-x)^3} = \sum_{k=1}^{\infty} k(k-1)x^{k-2} \quad \text{for } |x| < 1.$$

So

$$\begin{aligned} E[N^2] &= \sum_{k=1}^{\infty} k^2 P(N=k) \\ &= \sum_{k=1}^{\infty} k^2 p(1-p)^{k-1} \\ &= \sum_{k=1}^{\infty} (k^2 - k)p(1-p)^{k-1} + \sum_{k=1}^{\infty} kp(1-p)^{k-1} \\ &= p(1-p) \sum_{k=1}^{\infty} (k^2 - k)(1-p)^{k-2} + \sum_{k=1}^{\infty} kp(1-p)^{k-1} \\ &= p(1-p) \frac{2}{p^3} + \frac{1}{p} = \frac{2-2p+p}{p^2} = \frac{2-p}{p^2}. \end{aligned}$$

Therefore

$$\text{Var}(N) = E[N^2] - E[N]^2 = \frac{2-p}{p^2} - \frac{1}{p^2} = \frac{1-p}{p^2}.$$

This seems reasonable when you think about the case that $p = 1$ and what should happen as p gets small.

This material is a summary of Sections 2.1-2.4 from your textbook.

There are many additional examples and most of them will be quite useful as we move forward.